

MCP Ruby SDK: Unbounded line buffer in stdio transports leads to memory exhaustion (DoS)

Report Type: Weekly · Generated: July 30, 2026 17:59 UTC · Coverage: Jul 30 - Jul 30, 2026

1 Total Advisories	0 Critical	0 High	1 Medium
MEDIUM Severity	6.2 CVSS / Risk Score	TLP:CLEAR TLP Classification	PRO Customer Tier

Executive Summary

Summary The stdio transports in `MCP::Server::Transports::StdioTransport` and `MCP::Client::Stdio` read newline-delimited JSON-RPC frames using `IO#gets` with no `limit` argument. CRuby's `IO#gets` with no limit reads from the current position until the next separator (`\n`) with no upper bound on the returned string length. A peer that streams bytes without ever emitting a newline causes `gets` to accumulate the entire stream in a single Ruby `String` until the process is killed by the operating-system OOM killer. This is the same vulnerability class tracked in sibling MCP SDKs as GHSA-74gp-qhv5-v493 (Kotlin), GHSA-wqgc-pwpr-pq7r (TypeScript), GHSA-655q-2283-6jjj (Python), and others. It was identified during a cross-SDK audit; ruby-sdk had no prior report. ## Affected code Verified at `main` @ `cf44475c`. ### Server transport — [lib/mcp/server/transports/stdio_transport.rb](https://github.com/modelcontextprotocol/ruby-sdk/blob/cf44475c/lib/mcp/server/transports/stdio_transp

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Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
MCP Ruby SDK: Unbounded line buffer in stdio transports to neighbor	High	7.5	13.0%	## Summary The stdio transports in `MCP::Server::Transports::StdioTransport`

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
CDB-UNATTR-CVE	1	-

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intell/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

```
title: SENTINEL APEX - CDB-ADVISORY Exploitation Attempt
id: cdb-cdb-advisory-det
status: experimental
description: Detects exploitation indicators related to CDB-ADVISORY.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/30
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CDB-ADVISORY'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium
```

Microsoft Sentinel - KQL Query

```
// SENTINEL APEX - MCP Ruby SDK: Unbounded line buffer in stdio transports leads to memory exhaustion (DoS)
// Advisory | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "MCP Ruby SDK: Unbounded line buffer in s"
or AlertName has "MCP Ruby SDK: Unbounded line buffer in s"
or ExtendedProperties has "MCP Ruby SDK: Unbounded line buffer in s"
| extend Rule = "APEX-ADVISORY", Severity = "MEDIUM"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc
```

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield MCP Ruby SDK: Unbounded line buffer in stdio transports leads to memory exhaustion (DoS) instances exposed to the internet.

- 3. Enable verbose access logging on all MCP Ruby SDK: Unbounded line buffer in stdio transports leads to memory exhaustion (DoS) endpoints immediately.
- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for MCP Ruby SDK: Unbounded line buffer in stdio transports leads to memory exhaustion (DoS) or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$150K - \$900K Breach Cost Range (FAIR)	\$380K IBM Cost of Breach 2025 Median	\$120K/day Business Interruption Cost	\$85K Regulatory Fine Exposure
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Medium severity events require internal investigation and targeted customer notification. Average cost includes 340 hours of engineering time.

Remediation Cost Estimate: \$55K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
PCI-DSS v4.0	Req. 6.3.3	Non-critical patches applied within 3 months

Framework	Reference	Obligation
NIST CSF 2.0	ID.RA / PR.PS	Vulnerability assessment and platform security policies
ISO 27001:2022	A.8.8	Vulnerability management programme with documented remediation timelines

Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Monitor MCP Ruby SDK: Unbounded line buffer in stdio transports leads to memory exhaustion (DoS) for escalation.
2. Apply patches in next maintenance window.
3. Apply vendor patch for MCP Ruby SDK: Unbounded line buffer in stdio transports leads to memory exhaustion (DoS) immediately - check NVD for official fix guidance.
4. Enable enhanced logging and alerting on all systems running affected software versions.
5. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
6. Audit all internet-facing instances of affected products and confirm patch deployment.
7. Conduct a threat hunt using MITRE ATT&CK techniques mapped in this advisory.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--2b3d5641d36742ddd53cf18a
Report Type	Weekly
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
Generated At	2026-07-30T17:59:04.95381
Customer Tier	PRO
Customer Email	-
TLP	TLP: CLEAR
STIX Version	2.1
ATT&CK Version	v15

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